Ahmedabad University

CSE 641 Computer Vision

Weekly Report 3

Automated Segmentation of Post-Treatment Glioma Sub-regions and Resection Cavities from Multi-Parametric MRI.

Submitted to faculty: Prof. Mehul Raval

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Student Details

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Aim:

Conduct an Exploratory Data Analysis (EDA) on BraTS-GLI dataset to evaluate class imbalance, verify spatial structural integrity, and visualize tumor sub-regions across multi-planar MRI.

Introduction:

Before training models, it is crucial to understand the features of the dataset through EDA. The BraTS-GLI dataset contains multi-parametric MRI scans: T1, T1Gd, T2, and FLAIR modalities, as well as segmentation masks defining different tumor sub-regions such as Non-Enhancing Tumor Core (NETC), Surrounding Non-Enhancing FLAIR Hyperintensity (SNFH), Enhancing Tumor (ET), and Resection Cavity (RC).

EDA helps us to verify correctness of dataset and assures that all modalities are properly aligned for multi-channel input to models. It also helps us to analyze voxel intensity distributions, detect class imbalance among tumor regions, and understand tumor spatial characteristics within the brain volume. These results are essential for appropriate data preprocessing strategies like normalization, patch selection, and data augmentation before model training.

Work Completed:

We perform the following EDA on the BraTS-GLI Dataset:

1. **Structural Audit (Resolution, Shape, & Metadata):** We verifies the structural consistency of a particular patient from BraTS-GLI dataset by checking image shape, voxel resolution, datatype, and basic metadata of the MRI modalities and segmentation masks.

RESULT:

```
🔍 Found patient folder: BraTS-GLI-00005-100
🔍 Found 5 files inside.

============================================================
📊 HIGH-LEVEL STRUCTURAL AUDIT: BraTS-GLI-00005-100
============================================================
                 Modality/File Shape (H, W, D) Voxel Spacing (mm) Orientation Bit Depth  Isotropic
BraTS-GLI-00005-100-seg.nii.gz (182, 218, 182)    (1.0, 1.0, 1.0)         LAS   float32       True
BraTS-GLI-00005-100-t1c.nii.gz (182, 218, 182)    (1.0, 1.0, 1.0)         LAS   float32       True
BraTS-GLI-00005-100-t1n.nii.gz (182, 218, 182)    (1.0, 1.0, 1.0)         LAS   float32       True
BraTS-GLI-00005-100-t2f.nii.gz (182, 218, 182)    (1.0, 1.0, 1.0)         LAS   float32       True
BraTS-GLI-00005-100-t2w.nii.gz (182, 218, 182)    (1.0, 1.0, 1.0)         LAS   float32       True

--- Integrity Report ---
✅ 3D ALIGNMENT: Success. All volumes are co-registered spatially.
```

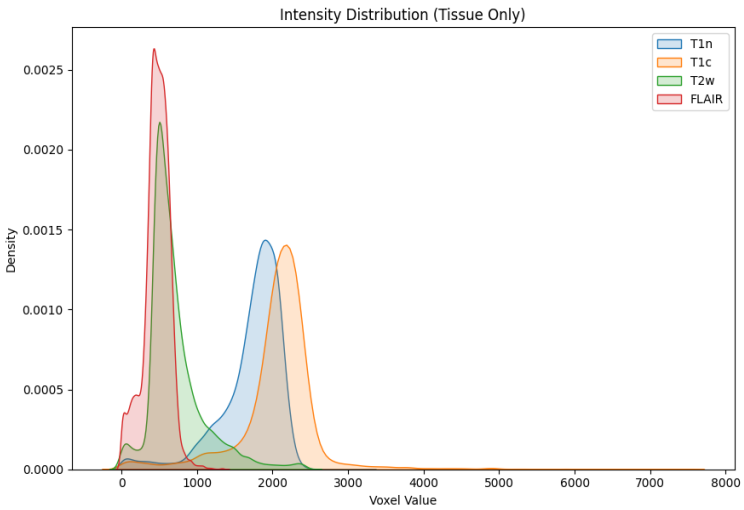
2. **Statistical & Intensity Analysis:** We analyzed that the intensity distribution of the four MRI modalities (T1n, T1c, T2w, and FLAIR) using non-zero brain voxels from a particular patient (BraTS-GLI-00005-100). Statistical measures such as mean, standard deviation, and 95th percentile were computed for each modality. The results showed observable variation in intensity ranges across modalities, as we can see in the figure below T1c having the highest mean intensity (2081.59) and FLAIR having the lowest (466.77). This confirms that each modality provides different imaging contrast and highlights the importance of applying intensity normalization. Also, clinical overlays verify perfect spatial alignment between the multi-class segmentation masks and the underlying anatomical structures, ensuring no spatial distortion.

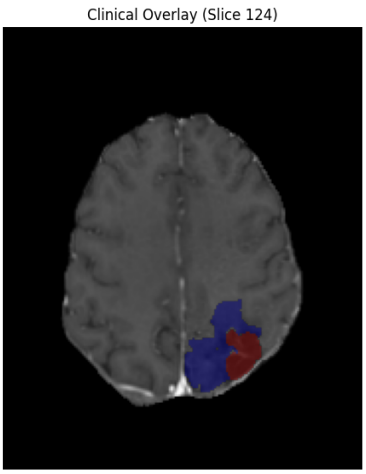
RESULT:

```
==================================================
🚀 HIGH-LEVEL EDA REPORT: BraTS-GLI-00005-100
==================================================

[1] Class Distribution (Target Imbalance)
           Label  Voxel Count  Volume %
              BG      7193826 99.623239
            SNFH        22960  0.317960
Resection Cavity         4246  0.058800

[2] Modality Intensity Profile
Modality        Mean        Std   95th Pctl
     T1n 1710.386316 424.239828 2164.880249
     T1c 2081.588960 615.876697 2603.728125
     T2w  709.044962 368.932366 1462.726624
   FLAIR  466.771259 169.486663  700.622205
```





3. **Clinical & Volumetric Analysis:** We analyzed the tumor burden and spatial extent of the lesion using the segmentation mask. Tumor sub-region presence, volumetric spread, slice-wise distribution, and tumor localization were examined through label-wise visualization. This is useful for understanding lesion extent and for planning segmentation-focused preprocessing.

RESULT:

Patient ID: BraTS-GLI-00005-100

```
ANALYSIS SUMMARY
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Total Tumor Volume:  27.21 cm³
Largest Region:      SNFH
Voxel Resolution:    1.0 x 1.0 x 1.0 mm
Center of Mass:      [109, 59, 124]
```
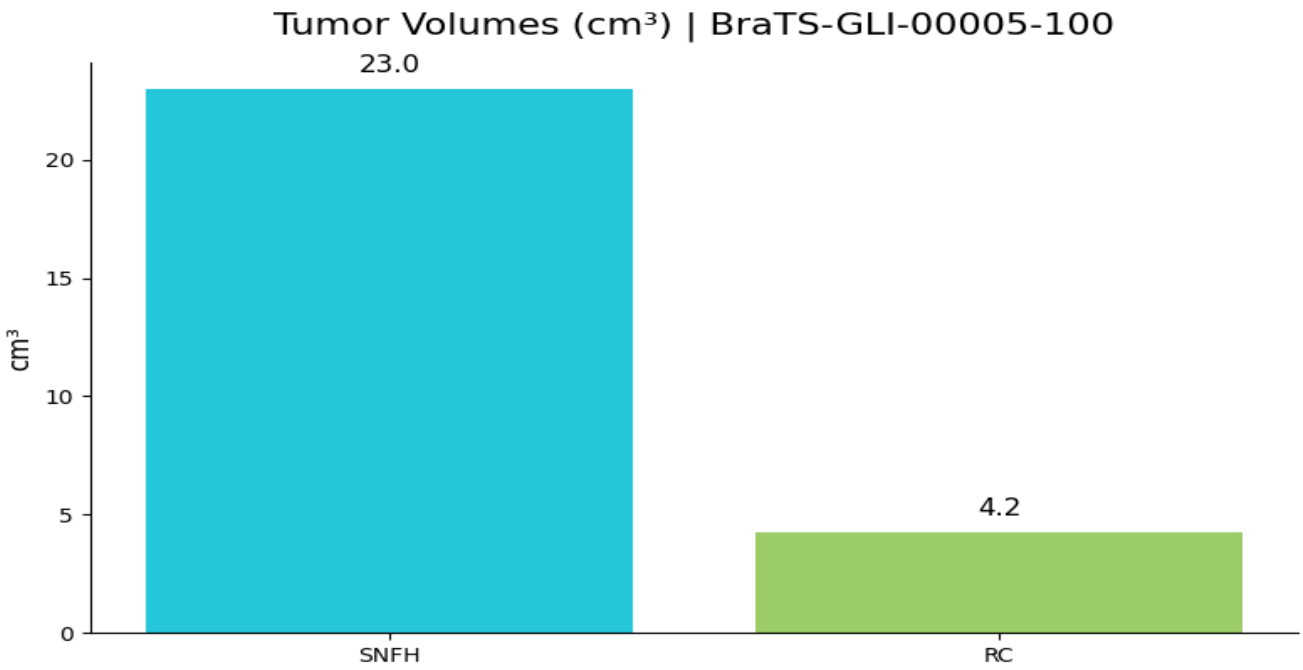
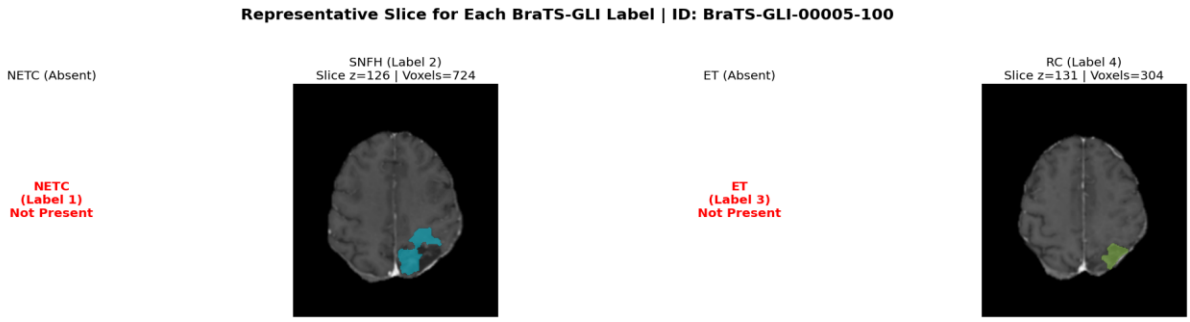
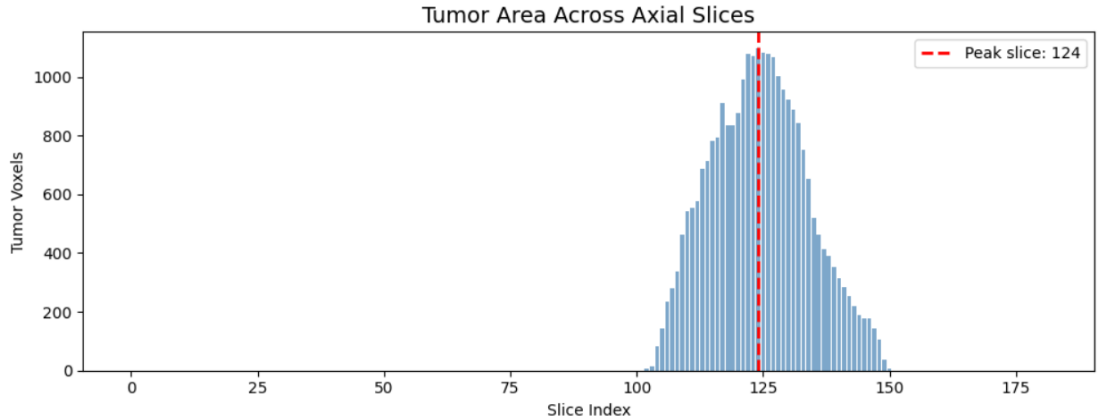





4. Slice wise Tumor Area: We analyzed how the tumor is actually spread across MRI volumns by plotting the number of pixels across every slice along the z-axis.This "slice-by-slice" plotting helped to show where exactly the tumor is more concentrated and distributed across the scan depth.So from this plot we found that the tumor only occupies a specific section of the brain rather than the whole volume.This is a key point as it helps to decide the depth of 3D patch size for model training.

RESULT:

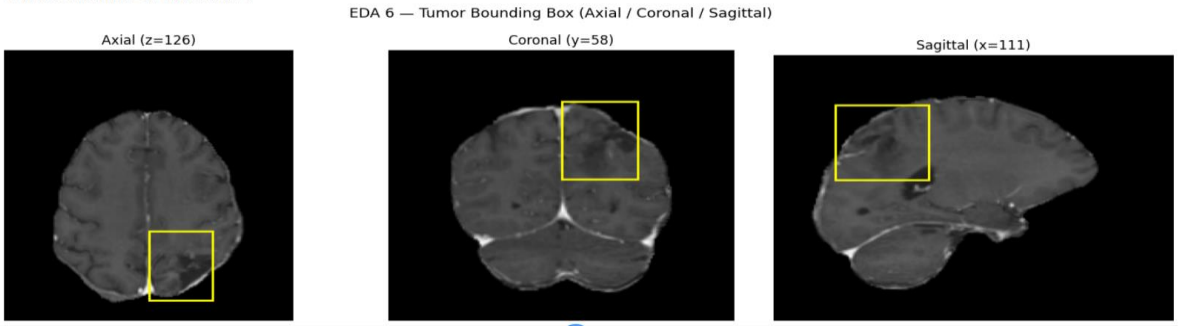
```
--- Slice-wise Tumor Distribution ---
  Total slices:        182
  Slices with tumor:   49 / 182
  Peak slice index:    124
  Max area (voxels):   1098
```



5. Bounding Box Analysis: We analyzed how much space the tumor takes up in the brain by calculating a 3D bounding box that tightly fits around all the tumor pixels. We visualized this on the T1c MRI slices by drawing a rectangle around the tumor and creating a zoomed-in view of the area with the segmentation overlay. By measuring the dimensions across the X, Y, and Z axes, we got a clear idea of the tumor's size.From these dimensions we derived the patch size for training by rounding up to the nearest power.

RESULT:

```
--- Tumor Bounding Box ---
  X: 91 → 131  (size: 40)
  Y: 33 → 84  (size: 51)
  Z: 102 → 150  (size: 48)

  Available modalities in images_data: 1
```



Next steps and goals:

- After completing the EDA, our next step is to build the data preprocessing pipeline for model training. This stage is important because the raw BraTS-GLI MRI scans are in NIfTI (.nii.gz) format, that cannot be directly used in models. The preprocessing pipeline will prepare the data in a structured format suitable for model training.
- This includes the following tasks:

    - **Setting up MONAI Transforms:** We will use MONAI dictionary transforms to resize, pad, and normalize the MRI images. We will define separate transform steps for the training and validation sets.
    - **Creating the Dataset:** To handle the heavy 3D MRI files efficiently, we will use monai.data.Dataset, to speed up the training time. We will also test using CacheDataset if our system memory allows it.
    - **Configuring PyTorch DataLoaders:** We will use PyTorch DataLoaders to handle the batching and shuffling of the data before giving it into the neural network.
    - **Validating Augmentations:** Before training the model, we will plot a random batch directly from the DataLoader. This is to check that our cropping, padding, and spatial augmentations are working correctly and not misinterpreting the brain anatomy.

Conclusion:

In conclusion, the EDA provided a practical and complete understanding of the BraTS-GLI dataset. By examin the structural metadata and intensity distributions, we confirmed that the four MRI modalities are perfectly aligned, but they will require strict intensity normalization before giving into a model. Furthermore, volumetric and bounding box analyses highlighted a severe class imbalance, proving that the actual tumor and resection cavity occupy only a very small fraction of the total brain volume. By looking at the tumors exact size and shape we chose the 3D patch size for training. This EDA phase worked well. Gave us the mathematical and

spatial details to create a good data preprocessing pipeline using MONAI for the next part of our project. We got the insights required to design an efficient pipeline. Gave us what we needed to target our MONAI data preprocessing pipeline. The results helped design an efficient MONAI data preprocessing.